

Project — Is Spike Counting Poisson?

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given 1,250 trials of spike times from one neuron at 5 different stimulus contrasts. Your job is to test whether the spike count on each trial follows a Poisson distribution, or whether the variance is bigger than the mean (overdispersion).

Background

For a Poisson process, the variance equals the mean:

$$\text{Var}(\text{spikes}) = \text{Mean}(\text{spikes}) \quad \rightarrow \quad \text{Fano factor} = \text{Var} / \text{Mean} = 1$$

If the Fano factor is much greater than 1, the variance is bigger than Poisson expects — there is extra 'noise' from the neuron itself.

Data

In `shared/data/`:

- `trial_protocol.csv` — `trial_id`, `contrast`, `role` (fit or holdout).
- `spike_times.csv` — one row per spike: `trial_id`, `contrast`, spike time.

Steps

1. Load and look. Open the starter notebook. Load both files. Print the number of trials per contrast.
2. Count spikes per trial. Group `spike_times` by `trial_id`. If a trial has zero spikes, count it as 0. Merge counts into the trial table.
3. Compute mean and variance by contrast. For each contrast, compute the mean spike count and the variance across trials. Compute the Fano factor (variance / mean).
4. Plot Fano vs. contrast. Plot Fano factor as a function of contrast. If $\text{Fano} > 1$, there is overdispersion.

What to hand in

- Your filled notebook.
- A table with contrast, mean spike count, variance, and Fano factor.
- The Fano-vs-contrast plot.

- A short paragraph (3-4 sentences) on whether the counts are Poisson.

If you finish early: also compute the Fano factor at a shorter counting window (e.g. first 400 ms of each trial). Does it change?